

1 Algebraic Structures

Definition 1.1 (Algebraic Structures). Let S be a set. A map $*$: $S \times S \rightarrow S$, $(s, t) \mapsto s * t$ is called a **binary operation**.

- $(S, *)$ is a **semigroup** if $*$ is associative:

$$\forall s, t, u \in S, \quad (s * t) * u = s * (t * u)$$

- A semigroup $(S, *)$ is a **monoid** if there exists a neutral element $e \in S$, i.e.,

$$\exists e \in S \quad \text{s.t.} \quad \forall s \in S: s * e = e * s = s$$

- A monoid $(S, *)$ is a **group** if every $s \in S$ has an inverse, i.e.,

$$\forall s \in S, \exists t \in S \quad \text{s.t.} \quad s * t = e = t * s$$

We call $(S, *)$ **commutative** if $\forall s, t \in S, s * t = t * s$. A commutative group is also called an **abelian group**.

Example 1.2. • $(\mathbb{N}, +) = (\{1, 2, 3, \dots\}, +)$ is a semigroup.

- $(\mathbb{Z}, +), (\mathbb{Q}, +), (\mathbb{R}, +), (\mathbb{C}, +)$ are abelian groups.
- $(\mathbb{N}_0, +) = (\{0, 1, 2, 3, \dots\}, +)$ is a monoid with neutral element 0.
- Let X be a set, then $(S = \text{Maps}(X, X) = \{f: X \rightarrow X\}, \circ)$ is a monoid with neutral element id_X .
- Let X be a set, then $(\mathcal{P}(X), \cap)$ is a monoid with neutral element X . Here $\mathcal{P}(X)$ is the power set of X and consists of all subsets of X .
- Let X be a set, $\text{Sym}(X) = \{f: X \rightarrow X \mid f \text{ is bijective}\}$, $(\text{Sym}(X), \circ)$ is a group, called the **symmetric group** of X . We write $S_n = \text{Sym}(\{1, \dots, n\})$.

Definition 1.3 (Rings). A **ring** is a triple $(R, +, \cdot)$, $+$ is addition, $\cdot$ is multiplication, such that:

1. $(R, +)$ is an abelian group with neutral element 0_R .
2. $(R, \cdot)$ is a monoid with neutral element 1_R .
3. The distributive laws hold: $\forall a, b, c \in R$,

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (b + c) \cdot a = b \cdot a + c \cdot a,$$

that is multiplication distributes over addition.

If $(R, \cdot)$ is commutative, we call $(R, +, \cdot)$ a **commutative ring**.

An element $r \in R$ is called a **zero divisor** if $r \neq 0_R$ and $\exists s \in R, s \neq 0_R$ such that $r \cdot s = 0_R$.

An element $r \in R$ is called a **unit** if r is invertible in $(R, \cdot)$, i.e., $\exists s \in R$ such that $r \cdot s = s \cdot r = 1_R$.

The set

$$R^* = \{r \in R \mid r \text{ is a unit}\}$$

is a group, called the **group of units** of R .

R is an **integral domain** if R is commutative, $1_R \neq 0_R$, and R has no zero divisors.

R is a **field** if $R^* = R \setminus \{0_R\}$ and R is commutative.

Example 1.4 (Rings, Zero Divisors, Units, and Fields).

• **Rings and Commutative Rings:**

1. $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, and $(\mathbb{C}, +, \cdot)$ are all commutative rings under ordinary addition and multiplication.
2. For $n \geq 1$, the ring of $n \times n$ matrices over real numbers, $(\text{Mat}(n \times n, \mathbb{R}), +, \cdot)$, is a non-commutative ring for $n \geq 2$.
3. For any integer $m \geq 2$, the ring of integers modulo m , $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$, is a commutative ring.

• **Zero Divisors:**

1. In $\mathbb{Z}/6\mathbb{Z}$, the elements $[2]$ and $[3]$ are zero divisors because $[2] \neq [0]$ and $[3] \neq [0]$, yet $[2] \cdot [3] = [6] = [0]$.
2. In $\text{Mat}(2 \times 2, \mathbb{R})$, non-zero matrices with determinant 0 are zero divisors. For instance:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

• **Units and Group of Units:**

1. In $(\mathbb{Z}, +, \cdot)$, the only elements with multiplicative inverses are 1 and -1 . Thus, the group of units is $\mathbb{Z}^* = \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$.
2. In the polynomial ring $\mathbb{R}[x]$, the units are precisely the non-zero constant polynomials, so $\mathbb{R}[x]^* = \mathbb{R} \setminus \{0\} = \mathbb{R}^*$.
3. For $\text{Mat}(n \times n, \mathbb{R})$, the group of units is the general linear group $\text{GL}(n, \mathbb{R}) = \{A \in \text{Mat}(n \times n, \mathbb{R}) \mid \det(A) \neq 0\}$.
4. In $\mathbb{Z}/m\mathbb{Z}$, the group of units consists of classes coprime to m : $(\mathbb{Z}/m\mathbb{Z})^* = \{[a] \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(a, m) = 1\}$.

• **Integral Domains:**

1. $\mathbb{Z}$ is an integral domain: it is commutative, $1 \neq 0$, and if $a \cdot b = 0$, then $a = 0$ or $b = 0$.
2. If R is an integral domain, then the polynomial ring $R[x]$ is also an integral domain.
3. $\mathbb{Z}/p\mathbb{Z}$ is an integral domain whenever p is a prime number.
4. $\mathbb{Z}/6\mathbb{Z}$ is *not* an integral domain due to the existence of zero divisors like $[2]$ and $[3]$.

• **Fields:**

1. $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are standard infinite fields.
2. For a prime p , $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is a finite field (also denoted $\mathbb{F}_p$).
3. $\mathbb{Z}$ is *not* a field because $2 \in \mathbb{Z} \setminus \{0\}$ lacks a multiplicative inverse in $\mathbb{Z}$.

Definition 1.5 (Modules and Vector Spaces). A **(left) module** over a ring $(R, +, \cdot)$ is an abelian group $(M, +)$ together with a map $R \times M \rightarrow M$, $(r, m) \mapsto r \cdot m$ (scalar multiplication) such that $\forall r, s \in R$ and $m, n \in M$:

1. $r \cdot (s \cdot m) = (r \cdot s) \cdot m$
2. $r \cdot (m + n) = r \cdot m + r \cdot n$
3. $(r + s) \cdot m = r \cdot m + s \cdot m$

4. $1_R \cdot m = m$

A **vector space** is a module over a field.

Example 1.6 (Modules vs. Vector Spaces).

1. Examples of Modules that are NOT Vector Spaces:

- **Abelian Groups as $\mathbb{Z}$ -Modules:** Any abelian group $(G, +)$ is naturally a $\mathbb{Z}$ -module via scalar multiplication $n \cdot g = \underbrace{g + \cdots + g}_{n \text{ times}}$ for $n > 0$. Since $\mathbb{Z}$ is an integral domain and not a field, these are not vector spaces unless $G = \{0\}$.
 - $\mathbb{Z}/n\mathbb{Z}$ (for $n \geq 2$) is a finite $\mathbb{Z}$ -module with torsion elements (e.g., $n \cdot [1] = [0]$). Vector spaces over fields cannot have non-zero elements annihilated by non-zero scalars.
 - $\mathbb{Z}^k$ (for $k \geq 1$) is a free $\mathbb{Z}$ -module. Submodules of free modules do not always admit complementary submodules (e.g., $2\mathbb{Z} \subset \mathbb{Z}$ has no complementary submodule in $\mathbb{Z}$).
- **Ideals in a Non-Field Ring R :** Viewing R as an R -module over itself, any ideal $I \subsetneq R$ is a submodule.
 - In $R = \mathbb{R}[x]$, the ideal $I = (x) = \{x \cdot p(x) \mid p(x) \in \mathbb{R}[x]\}$ is an R -module. Since $\mathbb{R}[x]$ is not a field, I is an R -module but not a vector space.
- **Modules with Torsion over Polynomial Rings:** Let V be a real vector space and $T \in \text{End}(V)$ a linear operator. V becomes an $\mathbb{R}[x]$ -module via $p(x) \cdot v = p(T)(v)$. Since $\mathbb{R}[x]$ is a principal ideal domain and not a field, V is an $\mathbb{R}[x]$ -module.

2. Examples of Vector Spaces:

- **Euclidean Spaces:** $\mathbb{R}^n$ over $\mathbb{R}$ (or $\mathbb{C}^n$ over $\mathbb{C}$) with standard componentwise operations.
- **Function Spaces:** For any set X and field $\mathbb{K}$, the set of functions $\text{Maps}(X, \mathbb{K}) = \{f: X \rightarrow \mathbb{K}\}$ is a $\mathbb{K}$ -vector space under pointwise addition and scalar multiplication.
- **Polynomial Vector Spaces:** $\mathbb{K}[x]$ (or the subspace $\mathbb{K}_n[x]$ of polynomials of degree $\leq n$) over the field $\mathbb{K}$.
- **Matrix Spaces:** $\text{Mat}(m \times n, \mathbb{K})$ over $\mathbb{K}$ with standard entrywise addition and scalar multiplication.
- **Field Extensions:** If $F \subset E$ is a subfield (e.g., $\mathbb{Q} \subset \mathbb{R}$ or $\mathbb{R} \subset \mathbb{C}$), then E is naturally a vector space over F . For instance, $\mathbb{C}$ is a 2-dimensional real vector space with basis $\{1, i\}$, while $\mathbb{R}$ is an infinite-dimensional vector space over $\mathbb{Q}$.

1.1 Constructions

1. Grothendieck Construction: If $(M, *)$ is a commutative monoid then we can construct an abelian group as follows: On $M \times M$ consider the relation:

$$(m, n) \sim (\tilde{m}, \tilde{n}) \iff \exists t \in M: m * \tilde{n} * t = \tilde{m} * n * t$$

This is an equivalence relation.

1. **Reflexivity:** For any $(m, n) \in M \times M$, choose $t = e \in M$. Then:

$$m * n * e = m * n = m * n * e$$

Thus, $(m, n) \sim (m, n)$.

2. **Symmetry:** Assume $(m, n) \sim (\tilde{m}, \tilde{n})$. Then there exists $t \in M$ such that:

$$m * \tilde{n} * t = \tilde{m} * n * t$$

Since $*$ is commutative, this is equivalent to:

$$\tilde{m} * n * t = m * \tilde{n} * t$$

Thus, $(\tilde{m}, \tilde{n}) \sim (m, n)$.

3. **Transitivity:** Assume $(m_1, n_1) \sim (m_2, n_2)$ and $(m_2, n_2) \sim (m_3, n_3)$.

Then there exist elements $t_1, t_2 \in M$ such that:

$$m_1 * n_2 * t_1 = m_2 * n_1 * t_1 \tag{1}$$

$$m_2 * n_3 * t_2 = m_3 * n_2 * t_2 \tag{2}$$

Multiply equation (1) by $n_3 * t_2$ and equation (2) by $n_1 * t_1$:

$$(m_1 * n_2 * t_1) * (n_3 * t_2) = (m_2 * n_1 * t_1) * (n_3 * t_2)$$

Using commutativity and associativity, rewrite both sides:

$$(m_1 * n_3) * (n_2 * t_1 * t_2) = (m_2 * n_3 * t_2) * (n_1 * t_1)$$

Substitute $(m_2 * n_3 * t_2) = (m_3 * n_2 * t_2)$ into the right-hand side using equation (2):

$$(m_1 * n_3) * (n_2 * t_1 * t_2) = (m_3 * n_2 * t_2) * (n_1 * t_1) = (m_3 * n_1) * (n_2 * t_1 * t_2)$$

Setting $t = n_2 * t_1 * t_2 \in M$, we obtain:

$$m_1 * n_3 * t = m_3 * n_1 * t$$

Hence, $(m_1, n_1) \sim (m_3, n_3)$.

Therefore, $\sim$ is an equivalence relation on $M \times M$. Let $G(M) := (M \times M) / \sim$ be the set of equivalence classes $[(m, n)]$. Define:

$$[(m, n)] * [(\tilde{m}, \tilde{n})] = [(m * \tilde{m}, n * \tilde{n})]$$

Then $(G(M), *)$ is a group.

Example 1.7. $(\mathbb{N}, +) \rightsquigarrow (\mathbb{Z}, +)$. Notation: $[(m, n)] = m - n$.

2. Localization: Let $(R, +, \cdot)$ be a commutative ring and $S \subset R$ a multiplicative subset (i.e., $\forall s, t \in S \implies s \cdot t \in S$, and $1_R \in S$). On $R \times S$ consider the equivalence relation:

$$(r, s) \sim (\tilde{r}, \tilde{s}) \iff \exists t \in S: t \cdot (r \cdot \tilde{s} - \tilde{r} \cdot s) = 0$$

Then $S^{-1}R = (R \times S) / \sim$ is a ring with respect to:

$$\text{Eq. class of } (r, s) \longrightarrow \frac{r}{s}, \quad \frac{r}{s} + \frac{\tilde{r}}{\tilde{s}} = \frac{r\tilde{s} + \tilde{r}s}{s\tilde{s}}, \quad \frac{r}{s} \cdot \frac{\tilde{r}}{\tilde{s}} = \frac{r\tilde{r}}{s\tilde{s}}, \quad 0_{S^{-1}R} = \frac{0_R}{1_R}, \quad 1_{S^{-1}R} = \frac{1_R}{1_R}$$

Example 1.8. • If $S = R^*$, then $S^{-1}R = R$.

- If $S = R \setminus \{0\}$ and R is a domain, then $S^{-1}R = \text{Quot}(R)$ is the quotient field (e.g., $R = \mathbb{Z} \implies S^{-1}R = \mathbb{Q}$).
- $R = \mathbb{C}[x]$, $S = \{f \in \mathbb{C}[x] \mid f(0) \neq 0\} \implies S^{-1}R = \{\frac{f}{g} \mid f, g \in \mathbb{C}[x], g(0) \neq 0\}$, the ring of rational functions defined near 0.

1.2 Substructures and Ideals

Let $(G, *)$ be a group. $H \subset G$ is a **subgroup** if $e_G \in H$ and $(H, *|_{H \times H})$ is a group:

$$\iff \forall h_1, h_2 \in H, h_1 * h_2^{-1} \in H$$

A subgroup $H \subset G$ is **normal** if $g * H * g^{-1} = H$ for all $g \in G$.

Let $(R, +, \cdot)$ be a ring. $S \subset R$ is a **subring** if $1_R \in S$, $0_R \in S$, and S is closed under $+$ and $\cdot$.

An additive subgroup $I \subset R$ is a **left (resp. right) ideal** if $R \cdot I = I$ (resp. $I \cdot R = I$). If I is both a left and right ideal, we say I is a **two-sided ideal**.

Let M be an R -module. $N \subset M$ is a **submodule** if $N \subset (M, +)$ is a subgroup and N is closed under scalar multiplication.

Remark 1.9. • A (left) ideal $I \subset R$ is the same as a submodule if we view R as a (left) R -module.

- A $\mathbb{Z}$ -module is the same as an abelian group.

Example 1.10 (Subgroups and Subrings).

1. Examples of Subgroups:

- **Trivial and Proper Subgroups:** For any group $(G, *)$, $\{e_G\}$ (the trivial subgroup) and G itself are always subgroups.
- **Integer Multiples:** In $(\mathbb{Z}, +)$, for any fixed integer $n \in \mathbb{Z}$, the set $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ of multiples of n is a subgroup (e.g., $2\mathbb{Z}$, the even integers). In fact, every subgroup of $\mathbb{Z}$ is of this form.
- **Subgroups of Matrix Groups:** Inside the general linear group $\text{GL}(n, \mathbb{R})$ (invertible matrices under multiplication):
 - The special linear group $\text{SL}(n, \mathbb{R}) = \{A \in \text{GL}(n, \mathbb{R}) \mid \det(A) = 1\}$ is a normal subgroup.
 - The orthogonal group $\text{O}(n) = \{A \in \text{GL}(n, \mathbb{R}) \mid A^t A = I_n\}$ is a subgroup.
- **Alternating Group:** Inside the symmetric group S_n , the set of even permutations $A_n = \{\sigma \in S_n \mid \text{sgn}(\sigma) = 1\}$ is a normal subgroup of index 2.
- **Subgroup of Units:** $(\mathbb{Q} \setminus \{0\}, \cdot)$ is a subgroup of $(\mathbb{R} \setminus \{0\}, \cdot)$.

2. Examples of Subrings:

- **Standard Number Systems:** $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ is a chain of subrings (where each ring contains 1 and is closed under addition, subtraction, and multiplication).
- **Gaussian Integers:** The set $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is a subring of the field of complex numbers $\mathbb{C}$.
- **Polynomial Subrings:**
 - The ring of constant polynomials $\mathbb{R}$ is a subring of $\mathbb{R}[x]$.
 - For any $n \geq 1$, $\mathbb{R}[x^n]$ is a subring of $\mathbb{R}[x]$.
- **Subrings of Functions:** Inside the ring of all real-valued functions $\text{Maps}(\mathbb{R}, \mathbb{R})$ under pointwise addition and multiplication:

- $C^0(\mathbb{R}, \mathbb{R})$ (continuous functions) is a subring.
- $C^\infty(\mathbb{R}, \mathbb{R})$ (smooth functions) is a subring of $C^0(\mathbb{R}, \mathbb{R})$.
- **Matrix Subrings:** The set of upper triangular matrices $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ with $a, b, c \in \mathbb{R}$ forms a subring of $\text{Mat}(2 \times 2, \mathbb{R})$.
- **Important Distinctions (Subring vs. Ideal):**
 - $2\mathbb{Z} \subset \mathbb{Z}$ is an *ideal* of $\mathbb{Z}$, but it is **not** a subring of $\mathbb{Z}$ because $1 \notin 2\mathbb{Z}$.
 - $\mathbb{Z} \subset \mathbb{Q}$ is a *subring* of $\mathbb{Q}$, but it is **not** an ideal of $\mathbb{Q}$ (e.g., $\frac{1}{2} \cdot 1 = \frac{1}{2} \notin \mathbb{Z}$).

1.3 Homomorphisms

Let G, H be groups. A map $\varphi: G \rightarrow H$ is a **group homomorphism** if $\forall g, \tilde{g} \in G$:

$$\varphi(g * \tilde{g}) = \varphi(g) * \varphi(\tilde{g})$$

Let R, S be rings. A map $\varphi: R \rightarrow S$ is a **ring homomorphism** if $\forall r, \tilde{r} \in R$:

$$\varphi(r + \tilde{r}) = \varphi(r) + \varphi(\tilde{r}), \quad \varphi(r \cdot \tilde{r}) = \varphi(r) \cdot \varphi(\tilde{r}), \quad \varphi(1_R) = 1_S$$

Let M, N be R -modules. A map $\varphi: M \rightarrow N$ is a **module homomorphism** if $\forall m, \tilde{m} \in M, r \in R$:

$$\varphi(m + \tilde{m}) = \varphi(m) + \varphi(\tilde{m}), \quad \varphi(r \cdot m) = r \cdot \varphi(m)$$

If M, N are vector spaces over a field $\mathbb{K}$, we call φ a **$\mathbb{K}$ -linear map**.

If $G = H / R = S / M = N$, we call $\varphi: G \rightarrow G$ an **endomorphism**.

Proposition 1.11. For a group / ring / module homomorphism φ , the kernel $\ker(\varphi) = \varphi^{-1}(\{0\})$ is a normal subgroup / ideal / submodule.

Proof. Only for the group case we verify the properties. First of all notice that $\varphi(e) = e$. Let $n \in \ker(\varphi)$ and consider gng^{-1} . We see that

$$\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g^{-1}) = \varphi(g)\varphi(g)^{-1} = e.$$

Notice that we also used the fact that $\varphi(g^{-1}) = (\varphi(g))^{-1}$. □

If the maps above are, in addition, bijective then we get the corresponding notions of **isomorphism** which is the equivalence between two algebraic structures.

2 Linear Algebra

Let $\mathbb{K}$ be a field, V a $\mathbb{K}$ -vector space, and $\mathcal{F} = (v_i)_{i \in I}$ a family of elements of V .

Definition 2.1. • A **linear combination** of elements of $\mathcal{F}$ is an expression $\sum_{i \in I} \lambda_i v_i$ where $\lambda_i \in \mathbb{K}$, and $\lambda_i \neq 0$ for only finitely many $i \in I$.

- $\mathcal{F}$ is **linearly independent** if $\sum_{i \in I} \lambda_i v_i = 0 \implies \lambda_i = 0, \forall i \in I$.
- The subspace spanned by $\mathcal{F}$ is $\text{Span}(\mathcal{F}) = \{\text{linear combinations of elements of } \mathcal{F}\} = \bigcap_{U \subset V, \mathcal{F} \subset U} U$, the smallest subspace containing $\mathcal{F}$.

- $\mathcal{F}$ generates V if $\text{Span}(\mathcal{F}) = V$.
- A **basis** for V is a linearly independent family that generates V .

Theorem 2.2. • Every $\mathbb{K}$ -vector space has a basis (possibly infinite).

- Any two bases of V have the same cardinality, called the **dimension** of V , $\dim V$.
- Any generating family contains a basis, and any linearly independent family is contained in a basis.

Proof. We begin by proving the third statement, as the first statement follows directly from it as a special case.

1. Any linearly independent family is contained in a basis :

Let V be a $\mathbb{K}$ -vector space, and let $S \subset V$ be a linearly independent set. Consider the family of sets

$$\mathcal{P} = \{X \subset V \mid S \subset X \text{ and } X \text{ is linearly independent}\},$$

partially ordered by set inclusion $\subseteq$.

Since $S \in \mathcal{P}$, $\mathcal{P}$ is non-empty. Let $\mathcal{C} = \{X_i\}_{i \in I}$ be a non-empty chain (a totally ordered subset) in $\mathcal{P}$. Define $U = \bigcup_{i \in I} X_i$. We claim that U is an upper bound for $\mathcal{C}$ in $\mathcal{P}$:

- Clearly, $S \subset U$.
- To show U is linearly independent, suppose $\sum_{j=1}^k \lambda_j u_j = 0$ for $u_1, \dots, u_k \in U$ and $\lambda_j \in \mathbb{K}$. Since $\mathcal{C}$ is a chain, there exists a single index $i_0 \in I$ such that $\{u_1, \dots, u_k\} \subset X_{i_0}$. Because $X_{i_0} \in \mathcal{P}$, X_{i_0} is linearly independent, so $\lambda_1 = \dots = \lambda_k = 0$.

Thus, $U \in \mathcal{P}$. By Zorn's Lemma, $\mathcal{P}$ contains a maximal element B .

By construction, B is linearly independent and contains S . If $\text{Span}(B) \neq V$, there would exist $v \in V \setminus \text{Span}(B)$. Then $B \cup \{v\}$ would be a strictly larger linearly independent set containing S , contradicting the maximality of B . Thus, $\text{Span}(B) = V$, so B is a basis of V containing S .

2. Every $\mathbb{K}$ -vector space has a basis:

Applying the extension result above to the empty set $S = \emptyset$ (which is vacuously linearly independent), it follows immediately that every vector space V possesses a basis.

3. Any generating family contains a basis:

Let $G \subset V$ generate V . Consider the family $\mathcal{Q} = \{X \subset G \mid X \text{ is linearly independent}\}$, partially ordered by inclusion $\subseteq$. Since $\emptyset \in \mathcal{Q}$, $\mathcal{Q}$ is non-empty.

By an identical Zorn's Lemma argument as above, every chain in $\mathcal{Q}$ has an upper bound in $\mathcal{Q}$ given by its union. Thus, by Zorn's Lemma, $\mathcal{Q}$ has a maximal element $B \subset G$.

Since $B \in \mathcal{Q}$, B is linearly independent. For any $g \in G \setminus B$, the maximality of B implies $B \cup \{g\}$ is linearly dependent, so $g \in \text{Span}(B)$. Hence, $G \subset \text{Span}(B)$, which implies $V = \text{Span}(G) \subseteq \text{Span}(B)$. Thus, $B \subset G$ is a basis of V .

□

Proof of Well-Definedness of Dimension (Invariance of Cardinality). Let B_1 and B_2 be two bases of V . We show that $|B_1| = |B_2|$.

Case 1: B_1 (or B_2) is finite.

Suppose $B_1 = \{v_1, \dots, v_n\}$ is a finite basis of V containing n elements, and let B_2 be any other basis of V .

We first prove by induction on n that any linearly independent set $W \subset V$ contains at most n elements.

Base Step ($n = 0$): If $n = 0$, then $V = \{0\}$. The only linearly independent subset of V is the empty set $\emptyset$, which has 0 elements. Thus $|W| \leq 0$.

Inductive Step: Assume that any vector space spanned by $n - 1$ vectors cannot contain more than $n - 1$ linearly independent vectors.

Suppose $W = \{w_1, \dots, w_m\} \subset V$ is a linearly independent set. Since $B_1 = \{v_1, \dots, v_n\}$ spans V , each w_j can be written as a linear combination of $v_1, \dots, v_n$. In particular, for w_1 :

$$w_1 = \sum_{i=1}^n c_i v_i = c_1 v_1 + c_2 v_2 + \dots + c_n v_n \quad \text{for some } c_i \in \mathbb{K}.$$

Since $w_1 \neq 0$ (as W is linearly independent), at least one $c_i \neq 0$. Without loss of generality, reorder $v_1, \dots, v_n$ so that $c_1 \neq 0$. We can solve for v_1 :

$$v_1 = c_1^{-1} w_1 - \sum_{i=2}^n c_1^{-1} c_i v_i.$$

This shows that $v_1 \in \text{Span}(\{w_1, v_2, \dots, v_n\})$, which implies:

$$V = \text{Span}(B_1) = \text{Span}(\{w_1, v_2, \dots, v_n\}).$$

Now consider the quotient space $V / \text{Span}(\{w_1\})$ ([see the next section for quotient vector spaces](#)).

- On one hand, $V / \text{Span}(\{w_1\})$ is spanned by the $n - 1$ images $\{v_2 + \text{Span}(\{w_1\}), \dots, v_n + \text{Span}(\{w_1\})\}$.
- On the other hand, the $m - 1$ images $\{w_2 + \text{Span}(\{w_1\}), \dots, w_m + \text{Span}(\{w_1\})\}$ are linearly independent in $V / \text{Span}(\{w_1\})$.

By the induction hypothesis applied to $V / \text{Span}(\{w_1\})$, the number of linearly independent elements cannot exceed the number of spanning elements:

$$m - 1 \leq n - 1 \implies m \leq n.$$

This completes the induction.

Conclusion: Since B_2 is linearly independent, every finite subset $W \subset B_2$ must satisfy $|W| \leq n$. This implies that B_2 cannot be infinite, so B_2 is finite and $|B_2| \leq n = |B_1|$.

Reversing the roles of B_1 and B_2 (using B_2 as the finite basis) yields $|B_1| \leq |B_2|$. Hence, $|B_1| = |B_2|$.

Case 2: Both B_1 and B_2 are infinite.

For every vector $w \in B_2$, since B_1 generates V , w can be expressed as a finite linear combination of elements in B_1 . That is, there exists a finite subset $J_w \subset B_1$ such that $w \in \text{Span}(J_w)$.

Consider the union of all such finite subsets:

$$\bigcup_{w \in B_2} J_w \subset B_1.$$

If this union were a proper subset of B_1 , say omitting $v_0 \in B_1$, then $B_2 \subset \text{Span}(B_1 \setminus \{v_0\})$. But since B_2 spans V , this would mean $\text{Span}(B_1 \setminus \{v_0\}) = V$, contradicting the linear independence of B_1 . Therefore, we must have:

$$B_1 = \bigcup_{w \in B_2} J_w.$$

Using standard properties of infinite cardinalities (where a union of $|B_2|$ finite sets has cardinality at most $|B_2| \cdot \aleph_0 = |B_2|$):

$$|B_1| = \left| \bigcup_{w \in B_2} J_w \right| \leq \sum_{w \in B_2} |J_w| \leq |B_2| \cdot \aleph_0 = |B_2|.$$

By symmetry, reversing the roles of B_1 and B_2 gives $|B_2| \leq |B_1|$. By the Cantor-Bernstein-Schröder theorem, $|B_1| = |B_2|$. $\square$

Example 2.3. • X a set, $V = \{f: X \rightarrow \mathbb{K} \mid f(x) \neq 0 \text{ for finitely many } x\} \implies$ has basis $(\delta_x)_{x \in X}$ where $\delta_x(y) = 1$ if $x = y$ and 0 otherwise.

- M a topological space, then $C^0(M) = \{f: M \rightarrow \mathbb{R} \text{ continuous}\}$ is an $\mathbb{R}$ -vector space, but usually hard to give an explicit basis.
- If V is a $\mathbb{K}$ -vector space, $n = \dim V$, then a choice of basis $\mathcal{B} = (v_1, \dots, v_n)$ is “the same” as a choice of isomorphism $V \cong \mathbb{K}^n$ via $\Phi_{\mathcal{B}}: V \rightarrow \mathbb{K}^n, \sum_{i=1}^n \lambda^i v_i \mapsto \begin{pmatrix} \lambda^1 \\ \vdots \\ \lambda^n \end{pmatrix}$.

2.1 Quotients and Dimension Formula

Let $U \subset V$ be a subspace. Define the quotient space V/U where $v \sim \tilde{v} \iff v - \tilde{v} \in U$.

If $\Phi: V \rightarrow W$ is a linear map, we consider $\ker \Phi = \{v \in V \mid \Phi(v) = 0\}$ and $\text{im } \Phi = \{\Phi(v) \mid v \in V\}$. We have the first isomorphism theorem:

$$V/\ker \Phi \cong \text{im } \Phi$$

In particular, if $\dim V < \infty$, then $\dim V/U = \dim V - \dim U$, leading to the **dimension formula**:

$$\dim V = \dim \ker \Phi + \dim \text{im } \Phi \quad (\text{where } \dim \text{im } \Phi = \text{rank } \Phi)$$

This is also sometimes called the **rank-nullity formula**.

Proof of the First Isomorphism Theorem. Let V and W be $\mathbb{K}$ -vector spaces, and let $\Phi: V \rightarrow W$ be a $\mathbb{K}$ -linear map. For brevity, set $K = \ker \Phi$ and $I = \text{im } \Phi$.

Define a map $\bar{\Phi}: V/K \rightarrow I$ by

$$\bar{\Phi}(v + K) = \Phi(v) \quad \text{for all } v \in V.$$

We must verify four properties: well-definedness, linearity, injectivity, and surjectivity.

1. **Well-definedness:** Suppose $v + K = v' + K$ for some $v, v' \in V$. Then $v - v' \in K = \ker \Phi$, which means $\Phi(v - v') = 0$. By linearity of Φ ,

$$\Phi(v) - \Phi(v') = \Phi(v - v') = 0 \implies \Phi(v) = \Phi(v').$$

Thus, $\bar{\Phi}(v + K) = \bar{\Phi}(v' + K)$, so $\bar{\Phi}$ is well-defined.

2. **Linearity:** For any $u + K, v + K \in V/K$ and $\lambda, \mu \in \mathbb{K}$, using the vector space operations in V/K and the linearity of Φ , we have:

$$\begin{aligned} \bar{\Phi}(\lambda(u + K) + \mu(v + K)) &= \bar{\Phi}((\lambda u + \mu v) + K) \\ &= \Phi(\lambda u + \mu v) \\ &= \lambda \Phi(u) + \mu \Phi(v) \\ &= \lambda \bar{\Phi}(u + K) + \mu \bar{\Phi}(v + K). \end{aligned}$$

Hence, $\bar{\Phi}$ is linear.

3. **Injectivity:** Suppose $\bar{\Phi}(v+K) = 0_W$. By definition of $\bar{\Phi}$, this implies $\Phi(v) = 0$. Therefore $v \in \ker \Phi = K$, which means $v + K = 0 + K$ in V/K . Thus, $\ker \bar{\Phi} = \{0 + K\}$, so $\bar{\Phi}$ is injective.
4. **Surjectivity:** Let $w \in I = \text{im } \Phi$. By definition of the image, there exists some $v \in V$ such that $\Phi(v) = w$. Then $\bar{\Phi}(v + K) = \Phi(v) = w$. Thus, $\bar{\Phi}$ is surjective.

Since $\bar{\Phi}$ is a linear bijection, it is a vector space isomorphism, and we conclude that

$$V/\ker \Phi \cong \text{im } \Phi.$$

□

The natural quotient map $\pi: V \rightarrow V/U$ satisfies the following **universal mapping property**:

If $\Phi: V \rightarrow W$ is linear and $U \subset \ker \Phi$, then Φ factors uniquely as $\Phi = \varphi \circ \pi$ with $\varphi: V/U \rightarrow W$ linear (defined by $\varphi(v + U) = \Phi(v)$).

$$\begin{array}{ccc} V & \xrightarrow{\Phi} & W \\ \pi \downarrow & \nearrow \exists! \varphi & \\ V/U & & \end{array}$$

2.2 Direct Sums and Dual Spaces

For vector spaces V, W , the **direct sum** $V \oplus W$ is given by $V \times W$ with componentwise operations. If V, W are finite-dimensional, $\dim(V \oplus W) = \dim V + \dim W$. If $(v_i)_{i \in I}$ and $(w_j)_{j \in J}$ are bases of V and W , then $\{(v_i, 0), (0, w_j) \mid i \in I, j \in J\}$ forms a basis for $V \oplus W$.

We define $\text{Hom}(V, W) = \{\Phi: V \rightarrow W \text{ linear}\}$ is a vector space. The **dual space** is $V^* = \text{Hom}(V, \mathbb{K})$. If $\dim V = n$ with basis $(v_1, \dots, v_n)$, the **dual basis** $(\alpha^1, \dots, \alpha^n)$ of V^* is characterized by $\alpha^i(v_j) = \delta_j^i$ where δ_j^i is the Kronecker delta.

$V \cong V^*$ non-canonically (depends on basis), but $V \cong V^{**}$ via the canonical isomorphism $v \mapsto [\alpha \mapsto \alpha(v)]$.

Theorem 2.4 (Canonical Isomorphism with Double Dual). Let V be a finite-dimensional $\mathbb{K}$ -vector space. The evaluation map

$$\text{ev}: V \rightarrow V^{**}, \quad v \mapsto \text{ev}_v, \quad \text{where } \text{ev}_v(\alpha) = \alpha(v) \text{ for all } \alpha \in V^*$$

is a canonical isomorphism of vector spaces.

Proof. We divide the proof into three steps: checking linearity of ev_v for a fixed v , linearity of the assignment $v \mapsto \text{ev}_v$, and proving ev is a bijection.

1. **Well-definedness ($\text{ev}_v \in V^{**}$):** For a fixed $v \in V$, we must show $\text{ev}_v: V^* \rightarrow \mathbb{K}$ is linear. For any $\alpha, \beta \in V^*$ and $a, b \in \mathbb{K}$:

$$\text{ev}_v(a\alpha + b\beta) = (a\alpha + b\beta)(v) = a\alpha(v) + b\beta(v) = a\text{ev}_v(\alpha) + b\text{ev}_v(\beta).$$

Thus, $\text{ev}_v \in (V^*)^* = V^{**}$.

2. **Linearity of ev:** For any $u, v \in V$ and $a, b \in \mathbb{K}$, we check if $\text{ev}_{au+bv} = a \text{ev}_u + b \text{ev}_v$. Evaluating both sides on an arbitrary functional $\alpha \in V^*$:

$$\begin{aligned} \text{ev}_{au+bv}(\alpha) &= \alpha(au + bv) \\ &= a\alpha(u) + b\alpha(v) \quad (\text{by linearity of } \alpha) \\ &= a \text{ev}_u(\alpha) + b \text{ev}_v(\alpha) \\ &= (a \text{ev}_u + b \text{ev}_v)(\alpha). \end{aligned}$$

Since this holds for all $\alpha \in V^*$, $\text{ev}(au + bv) = a \text{ev}(u) + b \text{ev}(v)$, so ev is linear.

3. **Injectivity:** Suppose $v \in \ker(\text{ev})$. Then $\text{ev}_v = 0$ in V^{**} , meaning $\text{ev}_v(\alpha) = 0$ for all $\alpha \in V^*$, which implies $\alpha(v) = 0$ for all $\alpha \in V^*$.

Assume for contradiction that $v \neq 0$. Extend $\{v\}$ to a basis $\{v = v_1, v_2, \dots, v_n\}$ of V , and let $\{v_1^*, \dots, v_n^*\}$ be its dual basis. Then $v_1^*(v) = 1 \neq 0$, which contradicts $\alpha(v) = 0$ for all $\alpha \in V^*$.

Therefore, we must have $v = 0$, so $\ker(\text{ev}) = \{0\}$, proving that ev is injective.

4. **Surjectivity and Conclusion:** Since V is finite-dimensional, $\dim V^* = \dim V$, and hence:

$$\dim V^{**} = \dim V^* = \dim V.$$

Since $\text{ev}: V \rightarrow V^{**}$ is an injective linear map between vector spaces of equal finite dimension, it is automatically an isomorphism (by the Rank-Nullity Theorem).

□

Given $\Phi: V \rightarrow W$, the dual map $\Phi^*: W^* \rightarrow V^*$ is defined by $\Phi^*(\beta) = \beta \circ \Phi: V \rightarrow \mathbb{K}$. Note that $(\Phi \circ \Psi)^* = \Psi^* \circ \Phi^*$.

2.3 Matrix Representation of Linear Maps

Let V, W be finite-dimensional $\mathbb{K}$ -vector spaces with bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$. For a linear map $\Phi: V \rightarrow W$, write:

$$\Phi(v_j) = \sum_{i=1}^m t_{ij} w_i, \quad \text{for } t_{ij} \in \mathbb{K}$$

Then $T = (t_{ij}) \in \text{Mat}(m \times n, \mathbb{K})$ is the matrix of Φ . In handy notation: $\Phi(v_1, \dots, v_n) = (w_1, \dots, w_m)T$. That is, the j -th column of the matrix T = coefficients of $\Phi(v_j)$ w.r.t $(w_1, \dots, w_m)$.

Example 2.5 (Matrix Representation of a Linear Map). Let $V = \mathbb{R}_2[x]$ be the vector space of real polynomials of degree ≤ 2 , with ordered basis:

$$\mathcal{B} = (v_1, v_2, v_3) = (1, x, x^2)$$

Let $W = \mathbb{R}^2$ with standard ordered basis:

$$\mathcal{C} = (w_1, w_2) = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

Define the linear map $\Phi: \mathbb{R}_2[x] \rightarrow \mathbb{R}^2$ by evaluating a polynomial and its derivative at $x = 1$:

$$\Phi(p(x)) = \begin{pmatrix} p(1) \\ p'(1) \end{pmatrix}$$

1. Expressing images of basis elements in terms of $\mathcal{C}$:

$$\Phi(v_1) = \Phi(1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \cdot w_1 + 0 \cdot w_2$$

$$\Phi(v_2) = \Phi(x) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \cdot w_1 + 1 \cdot w_2$$

$$\Phi(v_3) = \Phi(x^2) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1 \cdot w_1 + 2 \cdot w_2$$

2. Constructing the matrix T : Placing the coefficients of $\Phi(v_j)$ with respect to (w_1, w_2) into the j -th column yields $T \in \text{Mat}(2 \times 3, \mathbb{R})$:

$$T = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

3. Verifying the handy row-vector notation: Using $\Phi(v_1, v_2, v_3) = (w_1, w_2)T$, we have:

$$\Phi(1, x, x^2) = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right)$$

Base Change: Let $\tilde{\mathcal{B}} = (\tilde{v}_1, \dots, \tilde{v}_n)$ and $\tilde{\mathcal{C}} = (\tilde{w}_1, \dots, \tilde{w}_m)$ be new bases. Let $(\tilde{v}_1, \dots, \tilde{v}_n) = (v_1, \dots, v_n)S$ with $S \in \text{Mat}(n \times n, \mathbb{K})$ invertible, and $(\tilde{w}_1, \dots, \tilde{w}_m) = (w_1, \dots, w_m)U$ with $U \in \text{Mat}(m \times m, \mathbb{K})$ invertible. Then the new matrix $\tilde{T}$ satisfies:

$$\tilde{T} = U^{-1}TS$$

Example 2.6 (Base Change for a Matrix Representation). Let $V = \mathbb{R}^2$ and $W = \mathbb{R}^2$, and let $\Phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map given by $\Phi(x, y) = (2x + y, x + 2y)$.

1. Initial Bases and Matrix Representation T : Consider the standard bases $\mathcal{B} = \mathcal{C} = (e_1, e_2)$ where $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Applying Φ to the basis vectors:

$$\Phi(e_1) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2e_1 + 1e_2, \quad \Phi(e_2) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1e_1 + 2e_2$$

So the matrix representation of Φ with respect to $(\mathcal{B}, \mathcal{C})$ is:

$$T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

2. New Bases and Transition Matrices S and U : Define new bases $\tilde{\mathcal{B}} = \tilde{\mathcal{C}} = (\tilde{v}_1, \tilde{v}_2)$ given by:

$$\tilde{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = e_1 + e_2, \quad \tilde{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = e_1 - e_2$$

Expressing the new basis in terms of the old basis yields:

$$(\tilde{v}_1, \tilde{v}_2) = (e_1, e_2)S \quad \text{where} \quad S = U = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The inverse transition matrix is:

$$U^{-1} = S^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$$

3. Direct Calculation of $\tilde{T}$ in the New Basis: Evaluate Φ on the new basis vectors $\tilde{\mathcal{B}}$:

$$\Phi(\tilde{v}_1) = \Phi(1, 1) = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\tilde{v}_1 + 0\tilde{v}_2, \quad \Phi(\tilde{v}_2) = \Phi(1, -1) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0\tilde{v}_1 + 1\tilde{v}_2$$

Thus, the new matrix is diagonal:

$$\tilde{T} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

4. Verification via the Change of Basis Formula $\tilde{T} = U^{-1}TS$: Multiplying the matrices explicitly:

$$\begin{aligned} U^{-1}TS &= \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 3/2 + 3/2 & 1/2 - 1/2 \\ 3/2 - 3/2 & 1/2 + 1/2 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

This matches $\tilde{T}$ exactly.

Example 2.7. Let $V = \mathbb{K}_n[x]$ (polynomials of degree $\leq n$) and $\Phi = \frac{d}{dx}: V \rightarrow V$. With respect to $\mathcal{B} = (1, x, x^2, \dots, x^n)$, the matrix representation is:

$$T_{\frac{d}{dx}} = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & n \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}$$

Remark 2.8. If $\mathcal{B}^*, \mathcal{C}^*$ are dual bases, then $\Phi^*: W^* \rightarrow V^*$ is represented by T^t (the transpose matrix). This is because

$$\Phi^* \beta^j(v_i) = \beta^j(\Phi(v_i)) = \beta^j\left(\sum_k t_{ki} w_k\right) = t_{ji}, \quad \text{i.e., } \Phi^* \beta^j = \sum_{i=1}^n t_{ji} \alpha^i = \sum_{i=1}^n (T^t)_{ij} \alpha^i.$$

The **trace** of an endomorphism $\Phi: V \rightarrow V$ is $\text{Tr}(\Phi) = \text{Tr}(T) = \sum_{i=1}^n t_{ii}$ w.r.t. to a given basis. Since $\text{Tr}(AB) = \text{Tr}(BA)$, $\text{Tr}(\Phi)$ is basis independent.

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3 Bilinear and Hermitian Forms

Let $\mathbb{K}$ be a field and $V_1, \dots, V_n, W$ be $\mathbb{K}$ -vector spaces. A map $\beta: V_1 \times \dots \times V_n \rightarrow W$ is **n -multilinear** if for all i , the map from $V_i \rightarrow W$ given by $v_i \mapsto \beta(v_1, \dots, v_i, \dots, v_n)$ is linear. If $W = \mathbb{K}$ and $V_1 = \dots = V_n = V$, then β is called an **n -multilinear form** on V .

Definition 3.1. A multilinear form $\beta: V \times \dots \times V \rightarrow \mathbb{K}$ is called:

- **Symmetric** if $\beta(v_1, \dots, v_n) = \beta(v_{\pi(1)}, \dots, v_{\pi(n)})$, $\forall \pi \in S_n$.
- **Alternating** if $\beta(v_1, \dots, v_n) = 0$ whenever $v_i = v_j$ for some $i \neq j$.

For $n = 2$, an alternating form is also called **skew-symmetric**.

Definition 3.2. A bilinear form $\beta: V \times V \rightarrow \mathbb{K}$ is **non-degenerate** if $\beta(v, w) = 0, \forall w \in V \implies v = 0$ (equivalently, $V \rightarrow V^*, v \mapsto \beta(v, \cdot)$ is injective).

- An **inner product** on a real vector space V is a symmetric, positive definite bilinear form $g: V \times V \rightarrow \mathbb{R}$ (positive definite means $g(v, v) \geq 0$ and $g(v, v) = 0 \iff v = 0$).
- A **symplectic form** on a $\mathbb{K}$ -vector space V is a skew-symmetric, non-degenerate bilinear form $\omega: V \times V \rightarrow \mathbb{K}$.
- For $\mathbb{K} = \mathbb{C}$, a **Hermitian inner product** is a non-degenerate sesquilinear form $h: V \times V \rightarrow \mathbb{C}$ (linear in second slot, antilinear in first) such that $h(w, v) = \overline{h(v, w)}$. Recall the sesquilinear means

$$h(ax + y, z) = ah(x, z) + h(y, z), h(x, ay + z) = \bar{a}h(x, y) + h(x, z).$$

Let $n = \dim V$. The **Gram matrix** of a bilinear form $\beta: V \times V \rightarrow \mathbb{K}$ with respect to a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V is given by

$$B = (\beta(v_i, v_j))_{i,j=1}^n \in \text{Mat}(n \times n, \mathbb{K}).$$

Then

$$\beta\left(\sum_{i=1}^n x^i v_i, \sum_{j=1}^n y^j v_j\right) = \sum_{i,j=1}^n B_{ij} x^i y^j = (x^1 \dots x^n) B \begin{pmatrix} y^1 \\ \vdots \\ y^n \end{pmatrix}.$$

- β is non-degenerate $\iff \det B \neq 0$.
- β is symmetric $\iff B^t = B$.

If $(\tilde{v}_1, \dots, \tilde{v}_n) = (v_1, \dots, v_n)T$ with $T \in \text{Mat}(n \times n, \mathbb{K})$ invertible, then

$$\tilde{B} = (\beta(\tilde{v}_i, \tilde{v}_j))_{i,j=1}^n = T^t B T.$$

Pullback: For a linear map $\varphi: W \rightarrow V$ and a bilinear form $\beta: V \times V \rightarrow \mathbb{K}$, the pullback $\varphi^* \beta: W \times W \rightarrow \mathbb{K}$ is defined by

$$(\varphi^* \beta)(w_1, w_2) = \beta(\varphi(w_1), \varphi(w_2)).$$

If $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ are inner product spaces, and $\varphi: W \rightarrow V$ satisfies $\varphi^* \langle \cdot, \cdot \rangle_V = \langle \cdot, \cdot \rangle_W$, we call φ an **isometry**.

Theorem 3.3 (Sylvester's Law of Inertia). Let $\mathbb{K} = \mathbb{R}$ (or $\mathbb{C}$) and let $\beta: V \times V \rightarrow \mathbb{K}$ be a symmetric (or Hermitian) bilinear form. Then there exists a basis $(v_1, \dots, v_n)$ of V such that

$$B = \begin{pmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where the pair (p, q) (the signature) is uniquely determined by β .

In particular, if β is positive definite, then there exists an **orthonormal basis** $(v_1, \dots, v_n)$ of V , i.e., $\beta(v_i, v_j) = \delta_{ij}$.

Proof of Sylvester's Law of Inertia. We split the proof into two parts: the existence of such a diagonalizing basis and the uniqueness of the signature (p, q) .

Part 1: Existence (Gram-Schmidt / Orthogonal Diagonalization)

We proceed by induction on $n = \dim V$.

Base Step ($n = 1$): If $\dim V = 1$, let $v \in V \setminus \{0\}$.

- If $\beta(v, v) = 0$, then $B = (0)$, so $p = 0, q = 0$.
- If $\beta(v, v) > 0$, set $v_1 = \frac{v}{\sqrt{\beta(v, v)}}$. Then $\beta(v_1, v_1) = 1$, so $B = (1)$ with $p = 1, q = 0$.
- If $\beta(v, v) < 0$, set $v_1 = \frac{v}{\sqrt{-\beta(v, v)}}$. Then $\beta(v_1, v_1) = -1$, so $B = (-1)$ with $p = 0, q = 1$.

Inductive Step: Assume the claim holds for spaces of dimension $n - 1$.

- **Case 1:** $\beta(v, v) = 0$ for all $v \in V$.

By polarization identity, $\beta(u, v) = 0$ for all $u, v \in V$. Thus $\beta \equiv 0$, and the matrix representation is $B = 0_{n \times n}$ with respect to any basis. Here $p = 0, q = 0$.

- **Case 2:** There exists $w \in V$ such that $\beta(w, w) \neq 0$.

Scaling $w_1 = \frac{w}{\sqrt{|\beta(w, w)|}}$ ensures $\beta(w_1, w_1) = \pm 1$.

Consider the 1-dimensional subspace $W = \text{Span}(w_1)$ and its orthogonal complement:

$$W^\perp = \{v \in V \mid \beta(w_1, v) = 0\}.$$

The map $L: V \rightarrow \mathbb{K}$, $v \mapsto \beta(w_1, v)$ is a non-zero linear form (since $L(w_1) = \beta(w_1, w_1) \neq 0$). By rank-nullity, $\dim W^\perp = \ker L = n - 1$, and $V = W \oplus W^\perp$.

By the induction hypothesis applied to $\beta|_{W^\perp \times W^\perp}$, there exists a basis $(v_2, \dots, v_n)$ of $W^\perp$ with respect to which $\beta|_{W^\perp}$ has matrix representation $\text{diag}(I_{p'}, -I_{q'}, 0)$.

Reordering the basis $(w_1, v_2, \dots, v_n)$ so that vectors with $\beta(v_i, v_i) = 1$ come first, followed by -1 , and finally 0 , yields a basis $(v_1, \dots, v_n)$ of V where:

$$\beta(v_i, v_j) = \begin{cases} 1 & 1 \leq i = j \leq p, \\ -1 & p + 1 \leq i = j \leq p + q, \\ 0 & \text{otherwise.} \end{cases}$$

Part 2: Uniqueness of Signature (p, q)

Let $(v_1, \dots, v_n)$ and $(w_1, \dots, w_n)$ be two bases yielding matrices with signatures (p, q) and (p', q') , respectively.

Define the subspaces:

$$V_+ = \text{Span}(v_1, \dots, v_p), \quad V_- = \text{Span}(w_{p'+1}, \dots, w_{p'+q'}, w_{p'+q'+1}, \dots, w_n).$$

- For any non-zero $x \in V_+$, $\beta(x, x) > 0$.
- For any $y \in V_-$, $\beta(y, y) \leq 0$.

Therefore, $V_+ \cap V_- = \{0\}$. Applying the dimension formula for sum of subspaces:

$$\dim(V_+) + \dim(V_-) = \dim(V_+ \oplus V_-) \leq \dim V = n.$$

Substituting $\dim(V_+) = p$ and $\dim(V_-) = n - p'$ gives:

$$p + (n - p') \leq n \implies p \leq p'.$$

By symmetry, interchanging the roles of the two bases yields $p' \leq p$, so $p = p'$.

Since $\text{rank}(B) = p + q$ is an invariant of β ($\text{rank}(T^t B T) = \text{rank}(B)$ for invertible T), we also have $p + q = p' + q'$, which implies $q = q'$. Thus the signature (p, q) is unique.

Positive Definite Case

If β is positive definite, $\beta(v, v) > 0$ for all $v \neq 0$. Thus $q = 0$ and $p = n$, so $B = I_n$ and $\beta(v_i, v_j) = \delta_{ij}$. $\square$

If $\mathbb{K} = \mathbb{R}$ (or $\mathbb{C}$) and β is a (Hermitian) inner product, then we have the **Cauchy-Schwarz inequality**:

$$|\beta(v, w)|^2 \leq \beta(v, v)\beta(w, w).$$

If $v = \sum_{i=1}^n \lambda^i v_i$ and $(v_1, \dots, v_n)$ is an ONB, then $\lambda^i = \beta(v_i, v)$.

Examples:

- $\mathbb{R}^n$: $\langle v, w \rangle = \sum_{i=1}^n \lambda^i \mu^i$ for $v = \begin{pmatrix} \lambda^1 \\ \vdots \\ \lambda^n \end{pmatrix}, w = \begin{pmatrix} \mu^1 \\ \vdots \\ \mu^n \end{pmatrix}$.
- $\mathbb{C}^n$: $\langle v, w \rangle = \sum_{i=1}^n \bar{\lambda}^i \mu^i = (\bar{v}^t)(w)$. Then $g(v, w) = \operatorname{Re}(\langle v, w \rangle)$ is a real inner product, and $\omega(v, w) = -\operatorname{Im}(\langle v, w \rangle)$ is a symplectic form ($\mathbb{C}$ -linear in w).
- $C^0([a, b], \mathbb{C})$: $\langle f, g \rangle_{L^2} = \int_a^b \overline{f(t)} g(t) dt$.
- $V = \operatorname{Mat}(n \times m, \mathbb{C})$: $\langle A, B \rangle = \operatorname{tr}(\bar{A}^t B)$.
- $V = \mathbb{R}^4$: **Lorentz inner product**

$$\eta \left(\begin{pmatrix} x^0 \\ \vdots \\ x^3 \end{pmatrix}, \begin{pmatrix} y^0 \\ \vdots \\ y^3 \end{pmatrix} \right) = x^0 y^0 - \sum_{i=1}^3 x^i y^i.$$

If W is a real vector space, then $V = W \oplus W^*$ carries an indefinite inner product:

$$\langle (w, \alpha), (\tilde{w}, \tilde{\alpha}) \rangle = \frac{1}{2}(\alpha(\tilde{w}) + \tilde{\alpha}(w))$$

and a symplectic form:

$$\omega((w, \alpha), (\tilde{w}, \tilde{\alpha})) = \frac{1}{2}(\alpha(\tilde{w}) - \tilde{\alpha}(w)).$$

Fact: Any symplectic vector space is isomorphic to this model.

If V, W are finite-dimensional $\mathbb{K}$ ($\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$) vector spaces with (Hermitian) inner products $\langle \cdot, \cdot \rangle_V, \langle \cdot, \cdot \rangle_W$ and $\Phi: V \rightarrow W$ is linear, then there exists a unique **adjoint** $\Phi^\dagger: W \rightarrow V$ such that

$$\forall v \in V, w \in W: \quad \langle \Phi v, w \rangle_W = \langle v, \Phi^\dagger w \rangle_V.$$

We call an endomorphism $\Phi: V \rightarrow V$:

- **self-adjoint** / Hermitian if $\Phi^\dagger = \Phi$.
- **skew-adjoint** if $\Phi^\dagger = -\Phi$.
- **unitary** / **orthogonal** (or isometric) if $\Phi^\dagger = \Phi^{-1}$.
- **normal** if $[\Phi, \Phi^\dagger] = 0$.

Remark 3.4 (Normal Operators). The bracket $[\Phi, \Phi^\dagger] := \Phi \circ \Phi^\dagger - \Phi^\dagger \circ \Phi$ denotes the **commutator** of Φ and its adjoint $\Phi^\dagger$. Thus, an endomorphism $\Phi \in \text{End}(V)$ is **normal** if it commutes with its adjoint:

$$[\Phi, \Phi^\dagger] = 0 \iff \Phi \circ \Phi^\dagger = \Phi^\dagger \circ \Phi.$$

In terms of matrices with respect to an orthonormal basis, this corresponds to $AA^\dagger = A^\dagger A$ (where $A^\dagger = \bar{A}^t$).

Key properties of normal operators:

- **Generality:** Self-adjoint ($\Phi^\dagger = \Phi$), skew-adjoint ($\Phi^\dagger = -\Phi$), and unitary ($\Phi^\dagger = \Phi^{-1}$) operators are all automatically normal.
- **Spectral Theorem (over $\mathbb{C}$):** An endomorphism $\Phi \in \text{End}(V)$ is unitarily diagonalisable (i.e., V admits an orthonormal basis of eigenvectors of Φ) if and only if Φ is normal.

W.r.t. ONBs $(v_1, \dots, v_n)$ and $(w_1, \dots, w_m)$ of V, W : if T is the matrix representing Φ , then $\bar{T}^t$ is the matrix representing $\Phi^\dagger$.

Remark 3.5. The skew-adjoint, resp. Hermitian endomorphisms of a vector space form a vector space.

Example: $\mathbb{C}^2$ with $\langle \cdot, \cdot \rangle_{\text{std}}$. Then $E_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $E_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $E_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $E_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ form a basis for the 4-dimensional real vector space of Hermitian 2×2 matrices (Pauli matrices + I).

Remark 3.6. In Quantum Mechanics, we have a unitary vector space $\mathcal{H}$ of states:

- **Observables** $\longleftrightarrow$ Hermitian operators (eigenvalues = outcomes of measurement).
- **Symmetries** $\longleftrightarrow$ Unitary operators (time evolution, rotations, translations, ...).

If X is Hermitian and U is unitary, then $UXU^{-1} = UXU^\dagger$ is again Hermitian.

—

4 Determinants, Eigenvalues, and Canonical Forms

The determinant

Let $\mathbb{K}$ be a field. Then there exists a unique map $\det: \text{Mat}(n \times n, \mathbb{K}) \rightarrow \mathbb{K}$ such that:

D1) $\det$ is linear in each row.

D2) $\det$ is alternating, i.e., $\det A = 0$ if two rows of A are equal.

D3) $\det(I_n) = 1$.

Moreover:

D4) $\det(\lambda A) = \lambda^n \det A$.

D5) $\det(AB) = \det A \cdot \det B$ (so $\det(A^{-1}) = \frac{1}{\det A}$).

D6) $\det \begin{pmatrix} \vdots \\ a_i - \lambda a_j \\ \vdots \end{pmatrix} = \det \begin{pmatrix} \vdots \\ a_i \\ \vdots \end{pmatrix}$.

D7) $\det \begin{pmatrix} A & * \\ 0 & B \end{pmatrix} = \det A \cdot \det B$, and more generally $\det \begin{pmatrix} A & C \\ 0 & B \end{pmatrix} = \det A \cdot \det B$.

D8) $\det(A^t) = \det A$.

D9) $\det A \neq 0 \iff A$ is invertible.

D10) Laplace expansion:

$$\det A = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(\tilde{A}_{ij}),$$

where $\tilde{A}_{ij}$ is the matrix obtained by erasing the i -th row and j -th column.

Explicit formula:

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i\sigma(i)}.$$

- $n = 2$: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

- $n = 3$: Sarrus' rule.

If $\Phi: V \rightarrow V$ is an endomorphism and $\dim V = n$, we may define $\det \Phi := \det T$, where T is the matrix representing Φ w.r.t. a basis $\{v_1, \dots, v_n\}$. This is well-defined due to **D5**).

Eigenvalues and diagonalisability

Let V be a $\mathbb{K}$ -vector space.

If $\Phi \in \operatorname{End}(V)$, $\lambda \in \mathbb{K}$ is an **eigenvalue** if $\exists v \in V$, $v \neq 0$ such that $\Phi(v) = \lambda v$. Equivalently, $\ker(\Phi - \lambda \operatorname{id}_V) \neq \{0\}$. v is called an **eigenvector** for λ .

Proposition 4.1. Let V be a finite-dimensional vector space over a field $\mathbb{K}$ with $\dim V = n$, and let $\Phi: V \rightarrow V$ be a linear operator. Then $\lambda \in \mathbb{K}$ is an eigenvalue of Φ if and only if $\det(\Phi - \lambda \operatorname{id}_V) = 0$.

Proof. ($\implies$) Assume $\lambda \in \mathbb{K}$ is an eigenvalue of Φ . By definition, there exists a non-zero vector $v \in V$ ($v \neq 0$) such that

$$\Phi(v) = \lambda v \iff (\Phi - \lambda \operatorname{id}_V)(v) = 0.$$

Thus, $v \in \ker(\Phi - \lambda \operatorname{id}_V)$. Since $v \neq 0$, the operator $\Phi - \lambda \operatorname{id}_V$ has a non-trivial kernel and is therefore non-injective. In finite dimensions, a non-injective linear endomorphism is non-invertible, which implies

$$\det(\Phi - \lambda \operatorname{id}_V) = 0.$$

($\impliedby$) Assume $\det(\Phi - \lambda \operatorname{id}_V) = 0$. This implies that the operator $T = \Phi - \lambda \operatorname{id}_V$ is non-invertible. By the Rank-Nullity Theorem ($\dim V = \dim \ker T + \dim \operatorname{im} T$), a linear operator on a finite-dimensional vector space is invertible if and only if it is injective.

Since T is not invertible, it is not injective, so $\ker(\Phi - \lambda \operatorname{id}_V) \neq \{0\}$. Hence, there exists a non-zero vector $v \in V \setminus \{0\}$ such that

$$(\Phi - \lambda \operatorname{id}_V)(v) = 0 \implies \Phi(v) = \lambda v,$$

which proves that λ is an eigenvalue of Φ . □

Define $\chi_\Phi(t) := \det(\Phi - t \text{id}_V) \in \mathbb{K}[t]$ is the **characteristic polynomial** of Φ .

If λ is an eigenvalue of Φ , then its **geometric multiplicity** is $\dim E_\lambda = \dim \ker(\Phi - \lambda \text{id}_V)$. Its **algebraic multiplicity** is the multiplicity of λ as a root of χ_Φ .

We call Φ **diagonalisable** if V admits a basis consisting of eigenvectors.

In such a basis, Φ is represented by a diagonal matrix

$$T = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

We have the following.

Theorem 4.2. Φ is diagonalisable $\iff$ for all eigenvalues we have:

$$\text{alg. mult.} = \text{geom. mult.}$$

In particular, if $\dim V = n$ and Φ has n distinct eigenvalues, then Φ is diagonalisable.

Theorem 4.3. If $\mathbb{K} = \mathbb{R}$ (or $\mathbb{C}$), V is euclidean (unitary), and Φ is symmetric (normal), then Φ is diagonalisable.

Examples: Algebraic vs. Geometric Multiplicity

Let $V = \mathbb{R}^n$ (or $\mathbb{C}^n$). For an operator represented by a matrix A and an eigenvalue λ :

- **Algebraic multiplicity** (a_λ): The multiplicity of λ as a root of the characteristic polynomial $\chi_A(t) = \det(A - tI)$.
- **Geometric multiplicity** (g_λ): The dimension of the eigenspace $E_\lambda = \ker(A - \lambda I)$.

In general, $1 \leq g_\lambda \leq a_\lambda$. When $g_\lambda < a_\lambda$, the operator fails to be diagonalisable.

Proposition 4.4. Let $\Phi: V \rightarrow V$ be a linear operator on an n -dimensional vector space V , and let λ_0 be an eigenvalue of Φ . Then the geometric multiplicity of λ_0 is less than or equal to its algebraic multiplicity:

$$\text{gm}(\lambda_0) \leq \text{am}(\lambda_0).$$

Proof. Let $g = \text{gm}(\lambda_0) = \dim \ker(\Phi - \lambda_0 \text{id}_V)$. Choose a basis $\{v_1, \dots, v_g\}$ of the eigenspace E_{λ_0} , and extend it to a basis $\mathcal{B} = \{v_1, \dots, v_g, v_{g+1}, \dots, v_n\}$ for V .

Since $\Phi(v_i) = \lambda_0 v_i$ for $i = 1, \dots, g$, the matrix representation of Φ with respect to $\mathcal{B}$ has the block matrix form

$$[\Phi]_{\mathcal{B}} = \begin{pmatrix} \lambda_0 I_g & A \\ 0 & C \end{pmatrix},$$

where I_g is the $g \times g$ identity matrix, A is a $g \times (n - g)$ matrix, and C is an $(n - g) \times (n - g)$ matrix.

The characteristic polynomial of Φ is then given by

$$P_\Phi(\lambda) = \det([\Phi]_{\mathcal{B}} - \lambda I_n) = \det \begin{pmatrix} (\lambda_0 - \lambda) I_g & A \\ 0 & C - \lambda I_{n-g} \end{pmatrix}.$$

Using the determinant property for block upper-triangular matrices, we obtain

$$P_\Phi(\lambda) = \det((\lambda_0 - \lambda) I_g) \cdot \det(C - \lambda I_{n-g}) = (\lambda_0 - \lambda)^g \cdot q(\lambda),$$

where $q(\lambda) = \det(C - \lambda I_{n-g})$.

Since $(\lambda_0 - \lambda)^g$ divides $P_\Phi(\lambda)$, the root λ_0 has multiplicity at least g in $P_\Phi(\lambda)$. By definition, the algebraic multiplicity $a = \text{am}(\lambda_0)$ is the exact power of $(\lambda - \lambda_0)$ dividing $P_\Phi(\lambda)$, whence $g \leq a$. $\square$

Example 1: The 2×2 Standard Jordan Block

Consider the matrix:

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

1. Characteristic Polynomial & Algebraic Multiplicity:

$$\chi_A(t) = \det(A - tI) = \det \begin{pmatrix} 2-t & 1 \\ 0 & 2-t \end{pmatrix} = (2-t)^2$$

The only eigenvalue is $\lambda = 2$, with ****algebraic multiplicity**** $a_2 = 2$.

2. Eigenspace & Geometric Multiplicity: We solve $(A - 2I)v = 0$:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies y = 0$$

Thus, the eigenspace is $E_2 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$, which gives a ****geometric multiplicity**** $g_2 = \dim E_2 = 1$.

Since $g_2 = 1 < 2 = a_2$, A is not diagonalisable.

Example 2: A 3×3 Nilpotent Matrix

Consider the upper triangular matrix:

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

1. Characteristic Polynomial & Algebraic Multiplicity:

$$\chi_B(t) = \det(B - tI) = -t^3$$

The only eigenvalue is $\lambda = 0$, with ****algebraic multiplicity**** $a_0 = 3$.

2. Eigenspace & Geometric Multiplicity: We compute $\ker(B - 0I) = \ker(B)$:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies y = 0 \text{ and } z = 0$$

Thus, $E_0 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$, yielding ****geometric multiplicity**** $g_0 = \dim E_0 = 1$.

Here, $g_0 = 1 < 3 = a_0$.

Example 3: Partial Defect ($1 < g_\lambda < a_\lambda$)

Consider the 4×4 matrix with two Jordan blocks for the same eigenvalue $\lambda = 3$:

$$C = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

1. Characteristic Polynomial & Algebraic Multiplicity:

$$\chi_C(t) = (3 - t)^4 \implies a_3 = 4$$

2. Eigenspace & Geometric Multiplicity: Solving $(C - 3I)v = 0$:

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \implies x_2 = 0 \text{ and } x_4 = 0$$

The eigenspace is $E_3 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$, so the ****geometric multiplicity**** is $g_3 = 2$.

Here, $g_3 = 2 < 4 = a_3$ (the geometric multiplicity corresponds directly to the number of Jordan blocks).

In particular: If $\mathbb{K} = \mathbb{C}$, then unitary, self-adjoint, and skew-adjoint endomorphisms are diagonalisable.

Proof of Theorem 4.1. Let V be a finite-dimensional vector space over $\mathbb{K}$ with $\dim V = n$, and let $\Phi \in \text{End}(V)$. Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of Φ in $\mathbb{K}$.

($\implies$) Suppose Φ is diagonalisable. Then there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V consisting of eigenvectors of Φ . The characteristic polynomial $\chi_\Phi(t) = \det(\Phi - t \text{id}_V)$ factors completely over $\mathbb{K}$ into linear factors:

$$\chi_\Phi(t) = (-1)^n \prod_{i=1}^k (t - \lambda_i)^{a_i},$$

where $a_i = \text{alg. mult.}(\lambda_i)$ for each $i = 1, \dots, k$. Thus, $\sum_{i=1}^k a_i = n$.

Let $E_{\lambda_i} = \ker(\Phi - \lambda_i \text{id}_V)$ be the eigenspace corresponding to λ_i , and $g_i = \dim E_{\lambda_i} = \text{geom. mult.}(\lambda_i)$. Since eigenspaces associated with distinct eigenvalues form a direct sum, we have

$$\dim \left(\bigoplus_{i=1}^k E_{\lambda_i} \right) = \sum_{i=1}^k g_i \leq n.$$

Because $\mathcal{B}$ consists of eigenvectors, every element in $\mathcal{B}$ belongs to some E_{λ_i} . Hence, $V = \bigoplus_{i=1}^k E_{\lambda_i}$, which implies $\sum_{i=1}^k g_i = n$. Since $1 \leq g_i \leq a_i$ holds for every i , $\sum_{i=1}^k g_i = \sum_{i=1}^k a_i = n$ forces

$$g_i = a_i \quad \forall i \in \{1, \dots, k\}.$$

($\Leftarrow$) Suppose $\chi_\Phi(t)$ splits over $\mathbb{K}$ and $g_i = a_i$ for all $i = 1, \dots, k$. For each eigenspace E_{λ_i} , select a basis $\mathcal{B}_i$. Since the sum $\bigoplus_{i=1}^k E_{\lambda_i}$ is direct, the union $\mathcal{B} = \bigcup_{i=1}^k \mathcal{B}_i$ is a linearly independent set in V . The number of elements in $\mathcal{B}$ is

$$\sum_{i=1}^k g_i = \sum_{i=1}^k a_i = n = \dim V.$$

Therefore, $\mathcal{B}$ is a basis of V consisting entirely of eigenvectors, so Φ is diagonalisable.

In particular: If Φ has n distinct eigenvalues $\lambda_1, \dots, \lambda_n$, then $k = n$. Since $1 \leq g_i \leq a_i$ for each i and $\sum_{i=1}^n a_i = n$, it follows immediately that $a_i = g_i = 1$ for all $i = 1, \dots, n$. Thus Φ is diagonalisable. $\square$

Proof of Theorem 2 (Spectral Theorem). We prove the complex case first ($\mathbb{K} = \mathbb{C}$, V unitary, Φ normal), from which the real symmetric case follows directly.

Part 1: Complex Case (Φ Normal $\implies$ Unitarily Diagonalisable)

We proceed by induction on $n = \dim V$.

Base step ($n = 1$): Every endomorphism on a 1-dimensional space is diagonalisable.

Inductive step: Assume the claim holds for spaces of dimension $n - 1$. Since $\mathbb{C}$ is algebraically closed, the characteristic polynomial $\chi_\Phi(t)$ has at least one root $\lambda \in \mathbb{C}$. Thus, there exists a non-zero eigenvector $v_1 \in V$ such that $\Phi(v_1) = \lambda v_1$. Normalizing v_1 , we may assume $\|v_1\| = 1$.

Lemma: If Φ is normal and $\Phi(v) = \lambda v$, then $\Phi^\dagger(v) = \bar{\lambda}v$.

Proof of Lemma. Notice that $(\Phi - \lambda \text{id}_V)^\dagger = \Phi^\dagger - \bar{\lambda} \text{id}_V$. Computing the norm:

$$0 = \|(\Phi - \lambda \text{id}_V)v\|^2 = \langle (\Phi - \lambda \text{id}_V)v, (\Phi - \lambda \text{id}_V)v \rangle = \langle (\Phi - \lambda \text{id}_V)^\dagger(\Phi - \lambda \text{id}_V)v, v \rangle.$$

Since Φ is normal, $[\Phi - \lambda \text{id}_V, (\Phi - \lambda \text{id}_V)^\dagger] = 0$, so we can commute the operators:

$$0 = \langle (\Phi - \lambda \text{id}_V)(\Phi - \lambda \text{id}_V)^\dagger v, v \rangle = \|(\Phi^\dagger - \bar{\lambda} \text{id}_V)v\|^2 \implies \Phi^\dagger(v) = \bar{\lambda}v.$$

□

Now consider $W = \text{span}(v_1)^\perp$. We show that W is invariant under Φ : Let $w \in W$, so $\langle v_1, w \rangle = 0$. Using the lemma:

$$\langle v_1, \Phi(w) \rangle = \langle \Phi^\dagger(v_1), w \rangle = \langle \bar{\lambda}v_1, w \rangle = \bar{\lambda}\langle v_1, w \rangle = 0 \implies \Phi(w) \in W.$$

By the same argument, W is invariant under $\Phi^\dagger$. Thus, the restriction $\Phi|_W: W \rightarrow W$ is a normal endomorphism on W .

Since $\dim W = n - 1$, by the induction hypothesis there exists an orthonormal basis $(v_2, \dots, v_n)$ of W consisting of eigenvectors of $\Phi|_W$. Then $(v_1, v_2, \dots, v_n)$ forms an orthonormal basis of V consisting of eigenvectors of Φ , so Φ is diagonalisable.

Part 2: Real Case ($\mathbb{K} = \mathbb{R}$, V Euclidean, Φ Symmetric)

Let $\Phi \in \text{End}(V)$ be symmetric ($\Phi^\dagger = \Phi$).

First, we show that all eigenvalues of Φ are real. Complexifying V to $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ and extending Φ complex-linearly, let $\lambda \in \mathbb{C}$ be an eigenvalue of Φ with non-zero eigenvector $v \in V_{\mathbb{C}}$. Then:

$$\lambda \langle v, v \rangle = \langle \Phi v, v \rangle = \langle v, \Phi^\dagger v \rangle = \langle v, \Phi v \rangle = \bar{\lambda} \langle v, v \rangle.$$

Since $v \neq 0$, $\langle v, v \rangle > 0$, so $\lambda = \bar{\lambda}$, meaning $\lambda \in \mathbb{R}$.

Since all roots of $\chi_\Phi(t)$ are real, $\chi_\Phi(t)$ splits completely over $\mathbb{R}$. Every real symmetric matrix is normal, so applying the same induction argument on $W = \text{span}(v_1)^\perp \subset V$ over $\mathbb{R}$ yields an orthonormal basis $(v_1, \dots, v_n)$ of V consisting of real eigenvectors. □

Jordan canonical form

A **Jordan block** $J_{\lambda, n} \in \text{Mat}(n \times n, \mathbb{K})$ is a matrix of the form:

$$J_{\lambda, n} = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix} \quad \text{for some } \lambda \in \mathbb{K}.$$

Theorem 4.5. If $\mathbb{K} = \mathbb{C}$, then for any $\Phi \in \text{End}(V)$ there exists a basis such that Φ is represented by a matrix in **Jordan canonical form**:

$$T = \begin{pmatrix} J_{\lambda_1, n_1} & & 0 \\ & \ddots & \\ 0 & & J_{\lambda_k, n_k} \end{pmatrix}.$$

In particular, $\Phi = D + N$ with $D \in \text{End}(V)$ diagonalisable, $N \in \text{End}(V)$ nilpotent, and $DN = ND$, i.e., $[D, N] = 0$.

Example: $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\chi_A(t) = t^2 - 1$. $\implies \lambda = 1, -1$ are eigenvalues.

$$E_1(A) = \text{span} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} \right), \quad E_{-1}(A) = \text{span} \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} \right).$$

For $\mathcal{B} = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right)$, A looks like $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Eigenvalues of Hermitian operators are real:

$$\lambda \langle v, v \rangle = \langle \Phi v, v \rangle = \langle v, \Phi v \rangle = \bar{\lambda} \langle v, v \rangle \implies \lambda = \bar{\lambda}.$$

Eigenvalues of unitary operators are of unit length:

$$\text{if } Uv = \lambda v, \text{ then } \langle v, v \rangle = \langle Uv, Uv \rangle = \langle \lambda v, \lambda v \rangle = |\lambda|^2 \langle v, v \rangle \implies |\lambda|^2 = 1.$$

—

Tensor product

Definition 4.6. Let V, W be $\mathbb{K}$ -vector spaces. A **tensor product** $V \otimes W$ is a $\mathbb{K}$ -vector space together with a bilinear map $\otimes: V \times W \rightarrow V \otimes W$, $(v, w) \mapsto v \otimes w$ such that for any $\mathbb{K}$ -vector space Z and any bilinear map $\beta: V \times W \rightarrow Z$, there exists a unique linear map $\tilde{\beta}: V \otimes W \rightarrow Z$ making the following diagram commute:

$$\begin{array}{ccc} V \times W & \xrightarrow{\otimes} & V \otimes W \\ \beta \searrow & & \downarrow \exists! \tilde{\beta} \\ & & Z \end{array}$$

Theorem 4.7. There always exists a tensor product $V \otimes W$, unique up to isomorphism. If $\dim V, \dim W < \infty$, then $\dim(V \otimes W) = \dim V \cdot \dim W$.

Proof. Existence: Let $(v_i)_{i \in I}$ and $(w_j)_{j \in J}$ be bases of V and W , respectively. Consider the vector space T with basis $(e_{ij})_{(i,j) \in I \times J}$ and define $\otimes: V \times W \rightarrow T$, $(v_i, w_j) \mapsto e_{ij} =: v_i \otimes w_j$, extended bilinearly.

Uniqueness: Let $V \otimes W$ and $V \otimes' W$ be two tensor products. Consider the universal diagrams:

$$\begin{array}{ccc} V \times W & \xrightarrow{\otimes} & V \otimes W \\ \otimes' \searrow & & \downarrow \exists! f \\ & & V \otimes' W \end{array} \quad \begin{array}{ccc} V \times W & \xrightarrow{\otimes'} & V \otimes' W \\ \otimes \searrow & & \downarrow \exists! g \\ & & V \otimes W \end{array}$$

On the one hand, $g \circ f = \text{id}_{V \otimes W}$; on the other hand, $f \circ g = \text{id}_{V \otimes' W}$ by uniqueness, so f and g are mutually inverse isomorphisms. $\square$

Examples with $\dim V, \dim W < \infty$:

- $V^* \otimes W \cong \text{Hom}(V, W)$ via $\alpha \otimes w \mapsto [\alpha \otimes w: v \mapsto \alpha(v)w]$.
- $V^* \otimes W^* \cong \text{Bilinear}(V \times W, \mathbb{K})$.
- $V \otimes W \cong W \otimes V$.

If $(v_i)_{i=1}^n$ and $(w_j)_{j=1}^m$ are bases of V, W , and $(\alpha^i)_{i=1}^n$ is the dual basis of V^* , then any $\Phi \in \text{Hom}(V, W)$ can be written as:

$$\Phi = \sum_{i=1}^n \sum_{j=1}^m t_{ji} \alpha^i \otimes w_j,$$

where $T = (t_{ji})$ is the matrix representing Φ .

Tensor algebra

The **tensor algebra** of V is defined by:

$$\mathcal{T}(V) := \bigoplus_{k \geq 0} V^{\otimes k}, \quad \text{where } V^{\otimes 0} = \mathbb{K}, \quad V^{\otimes k} = \underbrace{V \otimes \cdots \otimes V}_{k \text{ times}}.$$

The product is given by tensor multiplication:

$$(v_1 \otimes \cdots \otimes v_k) \otimes (v_{k+1} \otimes \cdots \otimes v_{k+\ell}) = v_1 \otimes \cdots \otimes v_{k+\ell}.$$

Inside $V^{\otimes k}$ (or $(V^*)^{\otimes k}$), we have the subspace $\bigwedge^k V^*$ of **alternating tensors**.

More intuitively, $\bigwedge^k V^* = \text{Alt}^k(V; \mathbb{K}) = \{\beta: \underbrace{V \times \cdots \times V}_{k \text{ times}} \rightarrow \mathbb{K} \mid \beta \text{ is } k\text{-multilinear and alternating}\}.$

On $\bigwedge^* V^* = \bigoplus_{k \geq 0} \bigwedge^k V^*$, we have the **wedge product** (or exterior product), given for $\alpha_1, \dots, \alpha_k \in V^*$ by:

$$(\alpha_1 \wedge \cdots \wedge \alpha_k)(v_1, \dots, v_k) = \text{alt}((\alpha_i(v_j))_{i,j=1}^k) = \det((\alpha_i(v_j))_{i,j=1}^k).$$

Remark 4.8. If $\omega \in \bigwedge^k V^*$ and $\eta \in \bigwedge^\ell V^*$, then $\omega \wedge \eta = (-1)^{k\ell}(\eta \wedge \omega)$.

If $(e_1, \dots, e_n)$ is a basis of V and $(x^1, \dots, x^n)$ is the dual basis of V^* , then:

$$\{x^{i_1} \wedge \cdots \wedge x^{i_k} \mid 1 \leq i_1 < \cdots < i_k \leq n\}$$

is a basis of $\bigwedge^k V^*$.

In particular, $\dim \bigwedge^k V^* = \binom{n}{k}$.

If $\Phi: V \rightarrow \tilde{V}$ and $\Psi: W \rightarrow \tilde{W}$ are linear maps, then there is a unique linear map:

$$\Phi \otimes \Psi: V \otimes W \rightarrow \tilde{V} \otimes \tilde{W}, \quad v \otimes w \mapsto \Phi(v) \otimes \Psi(w).$$

Example: Let X, Y be finite sets, $V = \text{Maps}(X, \mathbb{K})$, and $W = \text{Maps}(Y, \mathbb{K})$. Then we may identify $V \otimes W \cong \text{Maps}(X \times Y, \mathbb{K})$ via $(f \otimes g)(x, y) = f(x)g(y)$.

Note: The tensor product appears naturally in quantum mechanics in the construction of the Hilbert space of composite systems.

Inner product on $\bigwedge^k V$

If V has an inner product $\langle \cdot, \cdot \rangle_V$, it induces an inner product on $\bigwedge^k V$ via:

$$\langle v_1 \wedge \cdots \wedge v_k, w_1 \wedge \cdots \wedge w_k \rangle = \det((\langle v_i, w_j \rangle)_{i,j=1}^k).$$

Hodge *-operator

Let $\dim V = n$. Since $\dim(\bigwedge^n V) = 1$, fix a generator $\text{vol} \in \bigwedge^n V$ of unit length. Then the **Hodge *-operator** is an isomorphism:

$$*: \bigwedge^k V \rightarrow \bigwedge^{n-k} V,$$

characterized by $\alpha \wedge *\beta = \langle \alpha, \beta \rangle \text{vol}$ for all $\alpha, \beta \in \bigwedge^k V$.

For example, if $n = 3$, $\text{vol} = e_1 \wedge e_2 \wedge e_3$, and $(e_1, \dots, e_n)$ is an orthonormal basis, then:

$$*(e_{i_1} \wedge \dots \wedge e_{i_k}) = \text{sgn}(\sigma) e_{j_1} \wedge \dots \wedge e_{j_{n-k}},$$

where $(i_1, \dots, i_k, j_1, \dots, j_{n-k})$ is a permutation $\sigma \in S_n$.